Metals

Virtually any metal that isn't hardened can be turned on your lathe. This opens up steels, aluminium, brass, copper and any other metal soft enough to work with. After testing with your file as noted before, you can be assured your high-speed steel woodturning tools can be effective on these materials.

Whether you are using metals as part of your turning project, or as fixturing for use in workholding, you can turn and drill these materials quite readily on your wood lathe. You won't have the ridigity of the cutting tool that you would have on a metal lathe but you'll have the freedom of shapes and movement that are difficult on a metal lathe.

One of the beauties of dealing with metals is that you'll often be starting with round stock. This has two immediate advantages. First, workholding is very simple and straightforward by using a chuck or a collet. Coming in standard diameters, a collet of the proper size will securely hold the stock for turning as well as being able to be precisely remounted should it need to be removed. Perhaps a bigger advantage is that it starts out round, not requiring the effort to remove corners. It will be round and accurately sized as purchased. That doesn't mean you may not work with block materials but more often than not you'll have the advantage of starting with round stock.

Unless you're given the material, you'll need to purchase it from an industrial supply house, but it is also available by mail order and on the internet. My suggestion would be to do a bit of research rather than just picking at random. The machinability of metals is based on their alloy and how it has been tailored. Aluminium, for example, can be machined nicely, but it can also be a nightmare depending on which alloy you select. A conversation with your local supplier or careful research online will let you pick metals and their different alloys that are most workable for you. The proper selection will give you the most favourable experience. A wealth of information is readily available on the internet to help guide you in your selection.

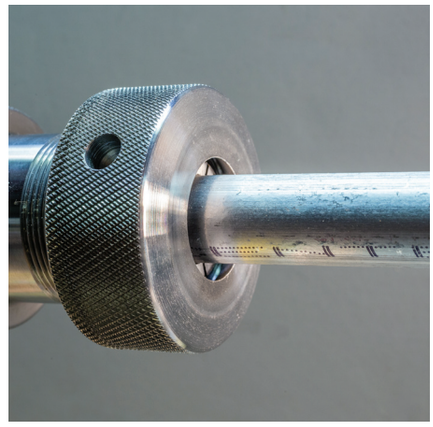
When purchasing metal bar stock, round or flat, it is usually sold in 3ft or 6ft lengths. You'll need to cut it to the appropriate length for use in your application. While a metal cutoff saw is a joy to which most won't have access, an inexpensive hack saw with the fine toothed metal blade will work. It takes more effort but it's very workable. While I've noted you can turn steels, I have two metal lathes so I do my steel fitting and adapter work on those. I use brass, aluminium and copper on my wood lathe to create finial, ferrule and pen parts.

Turning by hand allows for shaping and curves that are very difficult to execute on the metal lathe. Depending on your metal and the desired shape, you may find that scraping is superior to cutting. I can't provide any specific recommendation other than trial and error. Speeds and feeds will vary on the alloy as will the tool best to be used. I've been cautious to select free machining alloys of aluminium and brass to make my turning process as easy as possible.

When selecting your cutting tool, find a spot to test the performance of your cutting and scraping tools. I advise slow process. Because you'll be controlling the tool, you won't have the cutting control you would on a metal lathe. Too aggressive will be troublesome. You can use your standard woodturning abrasives to sand your metal turnings, especially if they are cloth-backed or wet/dry types, but if you intend to do much metal turning, you may wish to invest in emery cloth. Intended for metal work, you may appreciate it as an addition – or an alternative – to regular sandpaper abrasives.

If your town doesn't have a metals supplier, there are online sources for metals and plastics

A cutoff saw is a nicety but a regular hacksaw with a fine-tooth metal blade will cut your metal stock



The beauty of metals in round stock is the ability to workhold and accurately remount in a chuck or collet

A sharp, properly presented tool will deal with brass, aluminums, bronze and soft steels (alloy depending)

Seems like everyone should make at least one aluminum pen. Simple shapes work best with metals

A brass pull on a green turned burl lidded box. The brass cuts but also scrapes well

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